Passage–1

In India, while the unemployment rate is a frequently used measure of poor performance of the economy, under conditions of rising school and college enrolment, it paints an **inaccurate picture. The reported unemployment** rate is dominated by the experience of younger Indians who face higher employment challenges and exhibit greater willingness to wait for the right job than their older peers. The unemployment challenge is greater for people with secondary or higher education, and rising education levels inflate unemployment challenges.

71. Which one of the following statements most likely reflects as to what the author of the passage intends to say?

(a) Enrolment in schools and colleges is high but there is no quality education.
(b) Unemployment must be seen as a function of rising education and aspirations of young Indians.
(c) There are no labour-intensive industries to accommodate the huge number of unemployed people.
(d) The educations system should be properly designed so as to enable the educated people to be self-employed.

Ans. (b) : Statement that most likely reflects as to what the author of the passage intends to say is – "Unemployment must be seen as a function of rising education and aspirations of young Indians."

Passage–2

"Science by itself is not enough, there must be a force and discipline outside the sciences to coordinate them and point to a goal. It is not possible to run a course aright when the goal itself has not been rightly placed. What science needs is philosophy – the analysis of scientific method and the coordination of scientific purposes and results; without this, any science must be superficial. Government suffers, precisely like science, for lack of philosophy. Philosophy bears to science the same relationship which statesmanship bears to politics : movement guided by total knowledge and perspective, as against aimless and individual seeking. Just as the pursuit of knowledge becomes scholasticism when divorced from the actual needs of men and life, so the pursuit of politics becomes a destructive bedlam when divorced from science and philosophy."

72. Which one of the following statements best reflects the most rational, logical and practical message conveyed by the passage?

(a) Modern statesmen need to be well trained in scientific methods and philosophical thinking to enable them to have a better perspective of their roles, responsibilities and goals.
(b) It is not desirable to have Governments managed by empirical statesmen unless well mixed with others who are grounded in learning and reflect wisdom.
(c) As the statesmen/bureaucrats are the products of a society, it is desirable to have a system of education in a society that focuses on training its citizens in scientific method and philosophical thinking from a very early age.
(d) It is desirable that all scientists need to be philosophers as well to make their work goal-oriented and thus purposeful and useful to the society.

Ans. (a) : Statement that best reflects the most rational, logical and practical message conveyed by the passage is - Modern statesmen need to be well trained in scientific methods and philosophical thinking to enable them to have a better perspective of their roles, responsibilities and goals.

Passage–3

"The last end of the state is not to dominate men, nor to restrain them by fear; rather it is so to free each man from fear that he may live and act with full security and without injury to himself or his neighbour. The end of the state, I repeat, is not to make rational beings into bruke beasts and machines. It is to enable their bodies and their minds to function safely. It is to lead men to live by, and to exercise, a free reason; that they may not waste their strength in hatred, anger and guile, nor act unfairly toward one another."

73. Based on the above passage, which one of the following terms best expresses the ultimate goal of the state?

(a) Personal safety
(b) Health of body and mind
(c) Communal harmony
(d) Liberty

Ans. (d) : According to the passage, the term "Liberty" best expresses the ultimate goal of the state.

74. What is the remainder if 2^{192} is divided by 6?

(a) 0 (b) 1
(c) 2 (d) 4

Ans. (d) : According to the question,

$$\frac{2^{192}}{6}$$

Dividing 192 by 4, we get,

$$\frac{192}{4} \Rightarrow \text{Remainder} = 0$$

whenever remainder becomes 0, we put 4 as the power-

$$\frac{2^4}{6} = \frac{16}{6} \Rightarrow \text{Remainder} = 4$$

75. Consider the sequence
ABC_ _ ABC_DABBCD_ABCD
that follows a centain pattern.
Which one of the following completes the sequence?

(a) DACB (b) CDAB
(c) DCCA (d) DDCA

Ans. (d) : According to the given,
ABC _ _ ABC _ DABBCD _ ABCD
the following sequence can be writter as,
AB C D D / ABC C D / A B B CD / AA B C D
Therefore, DDCA will complete the sequence.

76. AB and CD are 2-digit numbers. Multiplying AB with CD results in a 3-digit number DEF. Adding DEF to another 3-digit number GHI results in 975. Further A, B, C, D, E, F, G, H, I are distinct digits. If E = 0, F = 8, then what is A + B + C equal to?

(a) 6 (b) 7
(c) 8 (d) 9

Ans. (a) : According to the given,

```
  D  E  F
  G  H  I
  9  7  5
```

Given E = 0, F = 8

```
  D  0  8
  G  H  I
  9  7  5
```

So, value of I = 7, H = 6
Now, the valve of D may be either 5 or 4
Case I:- When value of D is 4, then, AB × CD = 408
factors of 408 = 2 × 2 × 2 × 3 × 17
AB and CD are two digit numbers so the pairs formed are,
(i) 24 & 17 (ii) 34 & 12
Taking (ii) option, Therefore, A + B + C = 1 + 2 + 3
A + B + C = 6
Since no digits are same,
So value of A = 1, B = 2, C = 3.
Case II:- When value of D is 5
AB × CD = 508
factors of 508 = 2 × 2 × 127
Since it is a 3 digit number, this is incorrect.
Hence, A + B + C = 6

77. Consider the following statements in respect of five candidates P, Q, R, S and T. Two statements are true and one statement is false.
True Statement : One of P and Q was selected for the job.
False Statement : At least one of R and S was selected for the job.
True Statements : At most two of R, S and T were selected for the job.
Which of the following conclusions can be drawn?
1. At least four were selected for the job.
2. S was selected for the job.
Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : According to the given,

True statement →	One of P and Q was selected for the job.
False statement →	Atleast one of R or S got the job.
True Statement →	Maximum of two among R, S & T got the job.
Conclusion 1 →	Can not be determined
Conclusion 2→	Cannot be determined

78. Let P, Q, R, S and T be five statements such that :
I. If P is true, then both Q and S are true.
II. If R and S are true, then T is false.
Which of the following can be concluded?
1. If T is true, then at least one of P and R must be false.
2. If Q is true, then P is true.
Select the correct answer using the code given below :

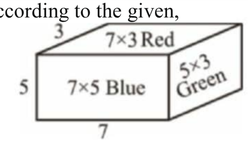
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : According to the given,
Statement I → If P is true, both Q and S are true.
Statement II → If R and S are true, then T is false.
Taking both statements into account,
Conclusion 1 is correct i.e. when T is true, then atleast P and R must be false.
Conclusion 2 can not be determined from both the statements because P is dependent on both Q & S.

79. A cuboid of dimensions 7 cm × 5 cm × 3 cm is painted red, green and blue colour on each pair of opposite faces of dimensions 7 cm × 5 cm, 5 cm × 3 cm, 7 cm × 3 cm respectively. Then the cuboid is cut and separated into various cubes each of side length 1 cm. Which of the following statements is/are correct?
1. There are exactly 15 cubes with no paint on any face.
2. There are exactly 6 small cubes with exactly two faces, one painted with blue and the other with green.
Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : According to the given,



Total volume of cuboids = 7 × 5 × 3
= 105 cm^3
Total number of cube of 1 cm^3 = 105
→ The (7 × 5) face is painted with blue and it has 3 layers. Therefore, 1 layer is inner layer.
→ The (7 × 3) face is painted with red and it has 5 layers. Therefore, 3 layers are inside layers.
→ The (5 × 3) face is painted with green and it has 7 layers. Therefore, 5 layers are inner layers.
So, total number of cubes with no painted face
= 5 × 3 × 1 = 15
Statement I is true.
→ Statement II is wrong.

80. The letters of the word "INCOMPREHENSIBILITIES" are arranged alphabetically in reverse order. How many positions of the letter/letters will remain unchanged?

(a) None (b) One
(c) Two (d) Three

Ans. (c) : According to the given,

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I N C O M P R E H E N S I B I L I T I E S
T S S R P O N N M L I I I I I H E E E C B
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Position of two letters will remain unchanged.